

Capstone Project Proposal

Title: Incident Investigation of Session Hijacking (stealing user's active login session) and Vendor Payment Fraud at the Ga Central Municipal District.

Track: Incident Investigation.

Target Organization: Ga Central Municipal District, a local government office managing public services like Property Rate Collection, utility data like Public Market Prepaid Metering, and regional infrastructure vendor contracts for Road Rehabilitation, Drainage Construction and the likes.

Problem Statement: Attackers successfully hijacked the active login session of the Municipal Finance Officer, Mrs. Yaa Afriyie, bypassing established authentication controls. Using this unauthorized access, the malicious actors intercepted and manipulated digital invoices to divert a public works vendor payment of GHS 450,000 intended for local road rehabilitation into a fraudulent bank account. This breach compromises the municipal office's financial integrity, threatens public funds, and exposes critical gaps in identity management and transaction verification protocols.

Objectives:

1. Identify the exact entry point and mechanism used by attackers to execute the session hijacking.
2. Trace and analyze unauthorized email account activity, log alterations, and configuration changes that led to a successful session hijacking
3. Assess the total financial and data exposure resulting from the compromised session.
4. Recommend specific, low-cost remediation steps, including secure email protocols and dual-authorization payment validation controls.

Scope: This project covers the investigation of the compromised manager's email account, related access logs, and the specific fraudulent transaction protocol. It explicitly excludes broader municipal network infrastructures, other departmental databases, or physical security controls.

Approach:

Week	Focus Point
2	reviewing network traffic and firewall log configurations to spot anomalous connection patterns.
3	detailed email log forensics and authentication header analysis to track the session hijacking.
5	incident response strategies, threat containment, and governance frameworks.
7	leverage AI tooling to analyze the incident timeline and automate policy documentation.

Expected Deliverable: A formal incident investigation report detailing the attack timeline, forensic evidence analysis, and localized threat containment strategies. Additionally, a concise slide deck will be prepared to present the technical findings and structural security recommendations on Demo Day.